Supplementary material for "Season-blocked validation, error structure and shortfall risk of soybean yield forecasts from satellite and reanalysis data in western Paraná, Brazil"

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S1 Predictors

Table S1: Predictors of the municipality–season dataset (93 in total). Phase predictors were computed for each of the four phases.

Predictor	Unit	Source	Phases
Precipitation	mm	CHIRPS	F1–F4
Days with $P < 1$ mm	days	CHIRPS	F1–F4
Potential evapotranspiration	mm	ERA5-Land	F1–F4
Actual evapotranspiration	mm	ERA5-Land	F1–F4
Water balance $P - ET_p$ (Eq. 1)	mm	CHIRPS, ERA5-Land	F1–F4
Days with $T_{max} > 32\,°C$	days	ERA5-Land	F1–F4
Mean air temperature	°C	ERA5-Land	F1–F4
Mean maximum air temperature	°C	ERA5-Land	F1–F4
Mean minimum air temperature	°C	ERA5-Land	F1–F4
Absolute maximum air temperature	°C	ERA5-Land	F1–F4
Mean dew-point temperature	°C	ERA5-Land	F1–F4
Mean soil temperature, 0–7 cm	°C	ERA5-Land	F1–F4
Volumetric soil water, 0–7 cm	$m^3\ m^{-3}$	ERA5-Land	F1–F4
Volumetric soil water, 7–28 cm	$m^3\ m^{-3}$	ERA5-Land	F1–F4
Global solar radiation	MJ m^{-2}	ERA5-Land	F1–F4
Net solar radiation	MJ m^{-2}	ERA5-Land	F1–F4
Leaf area index of low vegetation	–	ERA5-Land	F1–F4
Maximum NDVI	–	MOD13Q1, MYD13Q1	F1–F4
Mean NDVI	–	MOD13Q1, MYD13Q1	F1–F4
Maximum EVI	–	MOD13Q1, MYD13Q1	F1–F4
Mean EVI	–	MOD13Q1, MYD13Q1	F1–F4
Seasonal maximum NDVI and EVI	–	MOD13Q1, MYD13Q1	season
Number of valid vegetation observations	–	MOD13Q1, MYD13Q1	season
Soybean area	ha	MapBiomas Collection 10	season
Clay and sand content	%	Tluszcz (2025)	static
Latitude, longitude, altitude	°, m	IBGE	static

S2 Validation results

Table S2: Test performance of all datasets, validation schemes and models (MAE and RMSE in kg ha^{-1}).

Dataset	Scheme	Model	R^2	MAE	RMSE	R^2_{anom}	Skill
Earlier (replica)	Random	Trend	0.15	517	687	0.00	0.00
Earlier (replica)	Random	LightGBM	0.80	236	330	0.77	0.77
Earlier, misaligned	Random	Trend	0.15	518	685	0.00	0.00
Earlier, misaligned	LOSO	Trend	−0.03	565	755	0.00	0.00
Earlier, misaligned	Random	LightGBM	0.81	237	326	0.77	0.77
Earlier, misaligned	LOSO	LightGBM	−0.39	622	877	−0.35	−0.35
Earlier, aligned	Random	Trend	0.15	507	679	0.00	0.00
Earlier, aligned	LOSO	Trend	−0.02	554	747	0.00	0.00
Earlier, aligned	Random	LightGBM	0.81	226	317	0.78	0.78
Earlier, aligned	LOSO	LightGBM	0.25	477	640	0.27	0.27
New	Random	Trend	0.16	502	671	0.00	0.00
New	Random	Ridge	0.79	249	333	0.75	0.75
New	Random	Random forest	0.83	221	305	0.79	0.79
New	Random	XGBoost	0.82	227	313	0.78	0.78
New	Random	LightGBM	0.83	216	300	0.80	0.80
New	LOSO	Trend	0.02	547	735	0.00	0.00
New	LOSO	Ridge	0.59	361	475	0.58	0.58
New	LOSO	Random forest	0.45	411	550	0.44	0.44
New	LOSO	XGBoost	0.47	403	539	0.46	0.46

Dataset	Scheme	Model	R^2	MAE	RMSE	R^2_{anom}	Skill
New	LOSO	LightGBM	0.47	404	541	0.46	0.46
New	Rolling	Trend	−0.20	665	868	−0.01	0.00
New	Rolling	Ridge	0.37	500	627	0.47	0.48
New	Rolling	Random forest	0.22	544	699	0.34	0.35
New	Rolling	XGBoost	0.31	522	658	0.42	0.43
New	Rolling	LightGBM	0.27	524	675	0.39	0.40
New	Mun. out	Trend	0.60	338	468	0.00	0.00
New	Mun. out	LightGBM	0.79	245	337	0.48	0.48
New, + Landsat	LOSO	LightGBM	0.47	402	538	0.46	0.46

S3 Model and trend comparisons

Table S3: Differences in mean squared error between models under LOSO, with 95% intervals from 2000 bootstrap replicates of whole seasons.

Comparison (A − B)	ΔMSE (10^3 kg^2 ha^{-2})	95% interval	Replicates ≥ 0
Ridge - trend	−314.6	[−658.5; −95.4]	0.000
Ridge - RF	−77.4	[−166.4; 9.1]	0.043
Ridge - XGB	−64.6	[−140.8; 12.6]	0.050
Ridge - LGB	−67.6	[−145.5; 10.7]	0.051
LGB fusion - LGB MODIS only	−3.6	[−17.2; 8.6]	0.302

Table S4: Effect of the form of the technological trend under LOSO (MAE in kg ha^{-1}).

Trend	Model	R^2	MAE	Skill
Linear, municipal	Trend	0.02	547	0.000
Linear, municipal	Ridge	0.59	361	0.583
Quadratic, municipal	Trend	−0.04	567	0.000
Quadratic, municipal	Ridge	0.56	375	0.577
Linear, regional slope	Trend	0.00	549	0.000
Linear, regional slope	Ridge	0.60	363	0.596

S4 Performance by municipality

Table S5: LOSO performance of ridge regression by municipality (MAE and RMSE in kg ha^{-1}; AUC for shortfall below 80% of the trend, blank when no shortfall occurred).

Municipality	Microregion	MAE	RMSE	Skill	r anomaly	AUC shortfall
Anahy	Cascavel	236	281	0.65	0.85	1.00
Boa Vista da Aparecida	Cascavel	270	363	0.49	0.74	1.00
Braganey	Cascavel	246	326	0.35	0.74	1.00
Cafelândia	Cascavel	306	395	0.65	0.82	1.00
Campo Bonito	Cascavel	293	400	0.35	0.65	0.96
Capitão Leônidas Marques	Cascavel	406	556	−0.07	0.48	0.95
Cascavel	Cascavel	245	302	0.44	0.71	0.83
Catanduvas	Cascavel	277	392	−0.15	0.41	0.92
Corbélia	Cascavel	292	414	0.12	0.58	0.96
Diamante do Sul	Cascavel	367	433	0.28	0.57	1.00
Guaraniaçu	Cascavel	377	480	−0.52	0.34	0.96
Ibema	Cascavel	323	428	−0.09	0.48	0.83
Iguatu	Cascavel	260	410	−0.63	0.56	0.96
Lindoeste	Cascavel	321	414	0.06	0.56	0.98

Municipality	Microregion	MAE	RMSE	Skill	r anomaly	AUC shortfall
Nova Aurora	Cascavel	292	389	0.73	0.86	0.87
Santa Lúcia	Cascavel	432	582	−0.85	0.24	0.91
Santa Tereza do Oeste	Cascavel	282	377	0.04	0.57	
Três Barras do Paraná	Cascavel	348	470	0.10	0.51	0.96
Céu Azul	Foz do Iguaçu	279	371	0.69	0.84	1.00
Foz do Iguaçu	Foz do Iguaçu	525	706	0.39	0.63	0.86
Itaipulândia	Foz do Iguaçu	367	501	0.68	0.83	0.95
Matelândia	Foz do Iguaçu	402	496	0.65	0.81	0.97
Medianeira	Foz do Iguaçu	440	545	0.65	0.83	1.00
Missal	Foz do Iguaçu	437	579	0.63	0.80	0.95
Ramilândia	Foz do Iguaçu	327	426	0.69	0.84	0.97
Santa Terezinha de Itaipu	Foz do Iguaçu	387	532	0.55	0.75	0.92
Serranópolis do Iguaçu	Foz do Iguaçu	436	561	0.64	0.80	0.96
São Miguel do Iguaçu	Foz do Iguaçu	395	544	0.68	0.83	0.96
Vera Cruz do Oeste	Foz do Iguaçu	366	456	0.65	0.81	0.95
Assis Chateaubriand	Toledo	309	394	0.81	0.91	0.96
Diamante D'Oeste	Toledo	372	478	0.54	0.74	0.93
Entre Rios do Oeste	Toledo	432	526	0.65	0.84	0.95
Formosa do Oeste	Toledo	424	717	−0.36	0.38	0.82
Guaíra	Toledo	373	485	0.64	0.81	0.96
Iracema do Oeste	Toledo	370	432	0.75	0.87	0.96
Jesuítas	Toledo	322	373	0.79	0.89	0.96
Marechal Cândido Rondon	Toledo	387	461	0.72	0.86	0.96
Maripá	Toledo	380	493	0.71	0.85	0.93
Mercedes	Toledo	392	484	0.65	0.81	0.95
Nova Santa Rosa	Toledo	495	611	0.62	0.80	0.95
Ouro Verde do Oeste	Toledo	304	385	0.79	0.93	1.00
Palotina	Toledo	410	525	0.69	0.83	0.93
Pato Bragado	Toledo	453	533	0.65	0.85	0.98
Quatro Pontes	Toledo	476	555	0.60	0.78	0.90
Santa Helena	Toledo	408	508	0.64	0.81	0.94
São José das Palmeiras	Toledo	462	537	0.66	0.86	0.94
São Pedro do Iguaçu	Toledo	271	351	0.71	0.85	1.00
Toledo	Toledo	365	457	0.70	0.87	0.97
Tupãssi	Toledo	338	390	0.76	0.89	0.94

S5 Results by season

Table S6: Regional mean yield (kg ha^{-1}), Moran's I of the LOSO residuals and coverage (%) of the municipal 90% intervals by season.

Season	Observed	Forecast	Trend	Moran's I	p	Conformal	Mixed	CQR
2001	3239	3189	2763	0.02	0.248	96	98	98
2002	3118	2982	2815	0.26	0.001	100	100	100
2003	3134	3292	2841	0.21	0.001	98	98	98
2004	2386	2876	2953	0.71	0.001	71	76	53
2005	2623	2722	2944	0.08	0.086	98	98	92
2006	2239	2487	2994	0.31	0.002	84	84	63
2007	3070	3199	2947	0.21	0.004	100	100	100
2008	3271	3063	2957	0.26	0.002	100	100	100
2009	2182	2450	3041	0.22	0.002	76	76	88
2010	3319	3308	3001	0.40	0.001	100	100	100
2011	3428	3506	3019	0.31	0.002	100	100	100
2012	1931	2443	3104	0.34	0.001	71	71	55
2013	3553	3511	3057	0.42	0.001	100	100	100
2014	3437	3165	3082	0.31	0.001	98	98	98
2015	3365	3576	3106	0.55	0.001	98	98	100
2016	3473	3642	3121	0.00	0.356	100	100	100
2017	3735	3489	3126	0.46	0.001	96	98	96
2018	3596	3262	3151	0.25	0.002	90	90	94

Season	Observed	Forecast	Trend	Moran's I	p	Conformal	Mixed	CQR
2019	2694	2788	3234	0.18	0.010	96	96	100
2020	3947	3320	3157	0.39	0.001	61	63	69
2021	3413	4138	3221	0.55	0.001	49	53	96
2022	1634	1571	3443	0.26	0.001	80	80	47
2023	3279	3113	3278	0.27	0.001	92	92	96
2024	3243	3257	3307	0.33	0.001	100	100	100
2025	3610	3202	3267	0.47	0.001	82	82	94

S6 Shortfall risk

Table S7: Skill of shortfall probability forecasts by threshold (% of the trend).

Threshold	Frequency	Brier	Brier clim.	BSS	AUC
¡ 70%	0.117	0.058	0.107	0.46	0.96
¡ 80%	0.164	0.070	0.144	0.51	0.94
¡ 90%	0.243	0.094	0.194	0.52	0.92

Table S8: Reliability of shortfall probability forecasts: bins of forecast probability.

Threshold	Mean forecast probability	Observed frequency	Forecasts
¡ 70%	0.03	0.03	1072
¡ 70%	0.29	0.62	80
¡ 70%	0.49	0.85	20
¡ 70%	0.72	0.46	13
¡ 70%	0.96	0.88	40
¡ 80%	0.06	0.03	938
¡ 80%	0.29	0.28	112
¡ 80%	0.50	0.78	73
¡ 80%	0.67	0.82	44
¡ 80%	0.95	0.79	58
¡ 90%	0.09	0.02	651
¡ 90%	0.28	0.21	259
¡ 90%	0.49	0.45	93
¡ 90%	0.71	0.83	106
¡ 90%	0.92	0.86	116

S7 Spatial dependence of the residuals

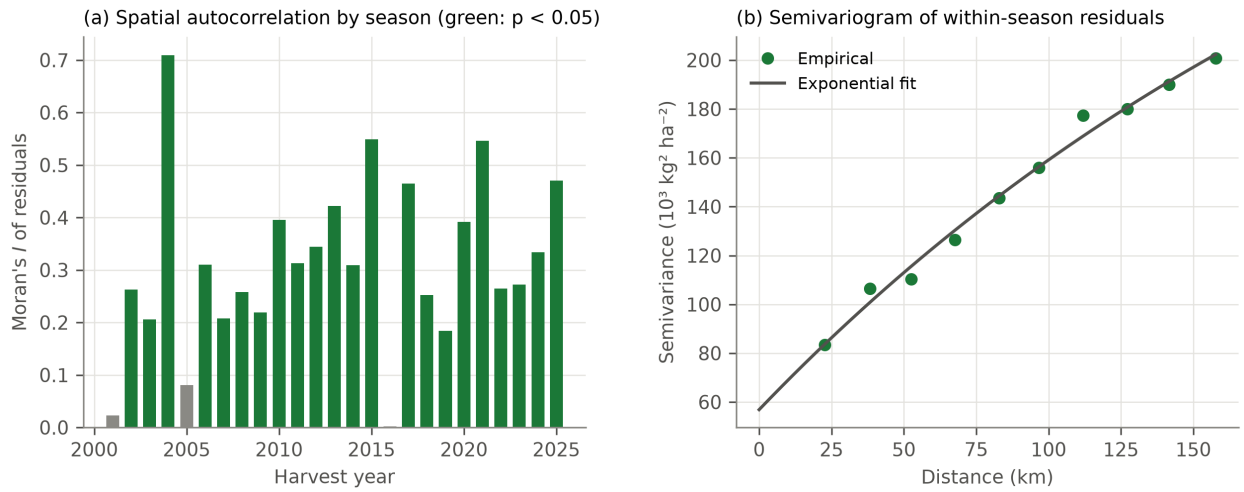


Figure S1: (a) Moran's I of the LOSO residuals of ridge regression by season, with six-nearest-neighbour weights and 999 permutations; (b) empirical semivariogram of the within-season residuals pooled over seasons, in 15-km distance classes with at least 20 pairs, and fitted exponential model.